

RESEARCH INTERESTS

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My research focuses on making AI systems genuinely *usable* for *diverse people*. I study post-training data and methods that explicitly optimize for human-AI ecosystems, and I develop human-grounded evaluations that are scalable and actionable. During my PhD, I have pursued two complementary directions toward this goal:

**Human-AI Interaction and Personalization:** I evaluate and improve how AI systems adapt to different users. My work explores new interaction paradigms, including multi-turn conversational agents and explanation-driven personalization, to better model users' backgrounds, needs, and preferences.

**Human-Centered Explanations:** I study explanations as a communication channel between AI systems and people. I investigate how models can produce high-quality explanations and how such explanations can improve human understanding, trust, and decision-making.

EDUCATION

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- **University of Southern California** Los Angeles, CA  
*Doctor of Philosophy, Computer Science* *Aug. 2021 – Apr. 2027 (expected)*
  - Advised By: [Xiang Ren](#) and [Swabha Swayamdipta](#)
  - [Apple Scholars in AI/ML PhD Fellowship](#), 2024-26
  - [Amazon ML PhD Fellow](#), 2023-24
  - USC Annenberg PhD Fellow, 2021-22
- **Indraprastha Institute of Information Technology** New Delhi, India  
*Bachelor of Technology, Computer Science and Engineering; CGPA: 9.30/10* *Aug. 2016 – Dec 2020*
  - Received the [Innovative Student Projects Award](#) for **best thesis in Computer Science** from the Indian National Academy of Engineering. One of the highest honors for undergraduates in India.
  - [Snap Research Scholarship](#), 2019

PUBLICATIONS AND PREPRINTS

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**Human-AI Interaction and Personalization**

- Priyanka Dey, **Brihi Joshi**, Preyashi Poddar, Jieyu Zhao, Emilio Ferrara. PALMs: Using Multi-Construct Grounded Rationales for Modeling Population Preferences in LLMs. *Under Submission*.
- Jacob Bremerman, **Brihi Joshi**, Hirona Arai, Xiang Ren, Jonathan May. Translating the Untranslatable: An Operationalizable Ontology for Untranslatability *Under Submission*.
- Natalie Mackraz, Andrew Silva, Aini Putkonen, **Brihi Joshi**, Joelle Alcaidinho, Arnav Arora, Barry-John Theobald, Katherine Metcalf. Evaluating the Evaluators: Investigating LLM Judges for Personalized Writing Style Assessment. *Under Submission*.
- Sahana Ramnath, Anurag Mudgil, **Brihi Joshi**, Skyler Hallinan, Xiang Ren. Amulet: Putting Complex Multi-Turn Conversations on the Stand with LLM Juries. *EMNLP 2025 (Main, Oral)*
- **Brihi Joshi**, Xiang Ren, Swabha Swayamdipta, Rik Koncel-Kedziorski, Tim Paek. Improving LLM Personas via Rationalization with Psychological Scaffolds. *EMNLP 2025 (Findings)*
- Rik Koncel-Kedziorski, **Brihi Joshi**, Tim Paek. PrimeX: A dataset of worldview, opinion, and explanation. *EMNLP 2025 (Main, Poster)*
- **Brihi Joshi\***, Keyu He\*, Sahana Ramnath, Sadra Sabouri, Kaitlyn Zhou, Souti Chattopadhyay, Swabha Swayamdipta, Xiang Ren. ELI-Why: Evaluating the Pedagogical Utility of Language Model Explanations. *ACL 2025 (Findings)*

- Jaspreet Ranjit, **Brihi Joshi**, Rebecca Dorn, Laura Petry, Olga Koumoundouros, Jayne Bottarini, Peichen Liu, Eric Rice, Swabha Swayamdipta. OATH-Frames: Characterizing Online Attitudes Towards Homelessness via LLM Assistants. *EMNLP 2025 (Main, Oral)*, **Outstanding Paper Award**

## Human-centered Explanations

- Keyu He, Tejas Srinivasan\*, **Brihi Joshi\***, Xiang Ren, Jesse Thomason, Swabha Swayamdipta. Believing without Seeing: Quality Scores for Contextualizing Vision-Language Model Explanations. *ACL 2026 (Main, Oral)*.
- Isabelle Lee, Emmy Liu, Cathy Jiao, Michael Saxon, **Brihi Joshi**, Dani Yogatama, Fazl Barez, Naomi Saphra. Rigorous Interpretation Is a Form of Evaluation. *Oral at EvalEval Workshop, ACL 2026*.
- **Brihi Joshi**, Sriram Venkatapathy, Mohit Bansal, Nanyun Peng, Haw-Shiuan Chang. CoKe: A Simple Yet Effective Approach to Story Evaluation via Chain-of-Keyword Rationalization. *Oral at GEM Workshop, ACL 2025*
- Sahana Ramnath, **Brihi Joshi**, Skyler Hallinan, Ximing Lu, Liunian Harold Li, Aaron Chan, Jack Hessel, Yejin Choi, Xiang Ren. Tailoring Self-Rationalizers with Multi-Reward Distillation. *ICLR 2024 (Poster)* and SetLLM Workshop at ICLR 2024.
- Aaron Chan\*, Zhiyuan Zeng\*, Wyatt Lake, **Brihi Joshi**, Hanjie Chen, Xiang Ren. KNIFE: Knowledge Distillation with Free-Text Rationales. *TrustML-(un)Limited Workshop at ICLR 2023*
- **Brihi Joshi\***, Ziyi Liu\*, Sahana Ramnath, Aaron Chan, Zhewei Tong, Shaoliang Nie, Qifan Wang, Yejin Choi and Xiang Ren. Are Machine Rationales (Not) Useful to Humans? Measuring and Improving Human Utility of Free-text Rationales. *ACL 2023 (Main, Oral)* and *Oral at trAI Workshop, CHI 2023*
- Dong-Ho Lee\*, Akshen Kadakia\*, **Brihi Joshi**, Aaron Chan, Ziyi Liu, Takashi Shibuya, Ryosuke Mitani, Toshiyuki Sekiya, Jay Pujara, Xiang Ren. XMD: An End-to-End Framework for Interactive Explanation-Based Debugging of NLP Models. *ACL Demo Track 2023*
- **Brihi Joshi\***, Aaron Chan\*, Ziyi Liu\*, Shaoliang Nie, Maziar Sanjabi, Hamed Firooz, Xiang Ren. ER-Test: Evaluating Explanation Regularization Methods for NLP Models. *EMNLP 2022 (Findings)* and *Oral at TrustNLP Workshop, NAACL 2022*

## Older Work

- **Brihi Joshi**, Neil Shah, Francesco Barbieri and Leonardo Neves. The Devil is in the Details: Evaluating Limitations of Transformer-based Methods for Granular Tasks. In *The 28th ACM International Conference on Computational Linguistics (COLING 2020)*.
- Aditya Chetan\*, **Brihi Joshi\***, Hridoy Sankar Dutta, Tanmoy Chakraborty. CoReRank: Ranking to Detect Users Involved in Blackmarket-based Collusive Retweeting Activities. In *The 12th ACM International Conference on Web Search and Data Mining (WSDM 2019)*. (Acceptance Rate: 16%, CORE2018 A\*)
- Udit Arora, Hridoy Sankar Dutta, **Brihi Joshi\***, Aditya Chetan\*, Tanmoy Chakraborty. Analyzing and Detecting Collusive Users Involved in Blackmarket Retweeting Activities. In *ACM Transactions on Intelligent Systems and Technology (TIST)*. (Impact Factor: **3.971**)
- **Brihi Joshi\***, Amogh Gulati\*, Chirag Jain\*, Jainendra Shukla. It's Not What They Play, It's What You Hear: Understanding Perceived vs. Induced Emotions in Hindustani Classical Music. In *22nd ACM International Conference on Multimodal Interaction, Late Breaking Reports (ICMI 2020)*.
- **Brihi Joshi\***, Shravika Mittal\*, Aditya Chetan\*. Did You "Read" the Next Episode? Using Textual Cues for Predicting Podcast Popularity. In *First Workshop on NLP for Music and Audio (NLP4MusA) at International Society for Music Information Retrieval Conference (ISMIR 2020)*.

- Hridoy Sankar Dutta, **Brihi Joshi\***, Aditya Chetan\*, Tanmoy Chakraborty. Retweet Us, We Will Retweet You: Spotting Collusive Retweeters Involved in Blackmarket Services. In *The 2018 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining (ASONAM 2018)*. (Acceptance Rate: 15%)
- Nishtha Madaan, Gautam Singh, Sameep Mehta, Aditya Chetan\*, **Brihi Joshi\***. Generating Clues for Gender based Occupation De-biasing in Text [arXiv:1804.03839](#) [cs.CL]

\* Equal Contribution

## INTERNSHIPS AND WORK EXPERIENCE

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- **Apple** Seattle, WA  
ML Research Intern, Apple AIML: Adaptive Devices and Platforms Team May 2025 – Aug 2025
- **Apple** Seattle, WA  
ML Research Intern, Apple AIML: Adaptive Devices and Platforms Team May 2024 – Dec 2024
- **Amazon** Cambridge, MA  
Applied Scientist II Intern, Alexa AI Natural Understanding: NLG Team May 2023 – Aug. 2023
- **Goldman Sachs** Bangalore, India  
Analyst (Full Time), Regulatory Engineering Team Dec. 2020 – July 2021
- **Snap Inc.** Los Angeles, CA  
Research Intern, Computational Social Science Team Sep. 2019 – Dec. 2019
- **Goldman Sachs** Bangalore, India  
Analyst Intern, Regulatory Engineering Team May 2019 - July 2019
- **IBM Research** New Delhi, India  
Research Intern, Fairness in ML Team May 2018 - July 2018

## AWARDS

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- **Apple Scholars in AL/ML PhD Fellowship**: Awarded for PhD research in Human Centred AI and Explainability.
- **Amazon ML PhD Fellowship**: Awarded for work in Explainable NLP, for AY 2023-24.
- **Outstanding Paper Award at EMNLP 2024**: Received for contribution to paper titled *OATH-Frames: Characterizing Online Attitudes Towards Homelessness via LLM Assistants* at EMNLP 2024.
- **Indian National Academy of Engineering Undergraduate Thesis Award 2021**: Awarded for research done as a part of undergraduate thesis in the Computer and Information Sciences Domain.
- **Annenberg Fellowship**: Awarded for admission to the PhD program at the University of Southern California.
- **Snap Research Scholarship 2019**: Awarded for research done in the field of Machine Learning. Award includes 10,000 USD and an offer to intern at Snap Research, USA. Only scholar from India!
- **AAAI 2020 Undergraduate Consortium**: Accepted to present my thesis at the AAAI 2020 Undergraduate consortium. Includes scholarship to attend to attend AAAI 2020
- **Microsoft Research India Travel Grant**: Awarded travel support of 50000 INR for visiting WSDM 2019
- **ACM-W Scholarship**: Awarded travel support of 1200 USD for visiting WSDM 2019
- **Google Women Techmakers Scholarship, 2018**: Awarded to students who work for diversity and inclusion in the field of Computer Science.
- **Best Technical Poster Runner-up at GHCI 2018**: Received for the project, “Generating Clues for Gender based Occupation De-biasing in Text”
- **Dean’s Award for Innovation R&D**: Awarded to students who work on Research projects beyond coursework. Awarded for the academic years 2016-17 and 2017-18.
- **Dean’s List of Academic Affairs**: Awarded to students who demonstrate excellence in an academic year. Awarded for the academic year 2016-17 and 2018-19.
- **Grace Hoppers Celebration India (GHCI) Scholarship, 2018**: Awarded travel grant and scholarship to attend the GHCI conference.

## OTHER ACADEMIC CONTRIBUTIONS

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- **Mentoring:**
  - Thor Christoffersen Hochman (Current USC CS Undergrad, CURVE Fellowship Fall 2025)
  - Harshavardhan Alimi (Current USC MS CS, **two research papers in progress**)
  - Keyu He (USC CS graduate, CURVE Fellowship Spring, Summer and Fall 2024, Spring 2025, Joined CMU MIIS in Fall 2025. Completed **two research papers**)
  - Anurag Mudgil (Current USC MS CS, Completed **one research paper.**)
  - Ziyi Liu (USC MS CS graduate, Joined USC CS PhD in Fall 2023. Completed **two research papers.**)
  - Zhewei Tong (Viterbi Tsinghua Undergraduate Summer Research Program, Joined CMU MS CS in Fall 2023. Completed **one research paper.**)
  - Pushpdeep Singh (Exchange student at USC via the Indo-US Science and Technology Forum).
- **Talks:**
  - Invited as a panelist on the [2025 Beyond the Ph.D. and Postdoctoral Conference](#) by the USC Career Centre, October 2025
  - Invited Talk at the [UCLA NLP Seminar](#), October 2025
  - Invited Talk at the [USC-Amazon Centre on Trusted AI](#) about Human-Centred Operationalization of Model Explanations, October 2023
- **Grant Writing:** *Utilizing Explanations for Model Refinement* in [Alexa: Fairness in AI 2022](#) award and writing contributions in multiple DARPA and IARPA project proposals under the supervision of Prof. Xiang Ren.
- **Reviewer:** EMNLP 2022, ACL 2023, EMNLP 2023, EMNLP 2024, ACL 2025, EMNLP 2025, ACL 2026
- **Co-Lead, [Women Who Code, Delhi](#)** Co-lead the Delhi Chapter of Women Who Code, a non-profit organization for upliftment of minorities in technology from 2017-2022.

## TEACHING EXPERIENCE

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|-----------------------------------------------------------------------------------------|----------------------------|
| • <b>CSCI 544: Graduate Applied Natural Language Processing</b>                         | USC                        |
| <i>Responsible for mentoring student projects and creating assignments</i>              | <i>Aug 2024 - Dec 2024</i> |
| • <b>CSE343: Machine Learning</b>                                                       | IIIT, Delhi                |
| <i>Teaching Assistant for a class of 150 senior undergraduate students</i>              | <i>Aug 2020 - Dec 2020</i> |
| • <b>CSE632: Semantic Web</b>                                                           | IIIT, Delhi                |
| <i>Teaching Assistant for a class of 170 senior undergraduate and graduate students</i> | <i>Jan 2020 - May 2020</i> |
| • <b>MTH201: Probability and Statistics</b>                                             | IIIT, Delhi                |
| <i>Teaching Assistant for a freshman class of 300 students</i>                          | <i>Jan 2019 - May 2019</i> |